

Dear Prof. Mikhasenko,

I hope this message finds you well.

I am Yijia Zeng, a PhD students in the Institute of High Energy physics in Beijing and now working for the BESIII experiment.

Recently, I am trying to perform an amplitude analysis to a three-body decay of J/psi with huge statistics. I strongly believe the method named "Dalitz-plot decomposition" proposed by your acticle PHYSICAL REVIEW D 101, 034033 (2020), provides the faster calculation and costs less memory than the other known approaches. Hence, I spent several weeks in coding a fit program based on your acticle in C++.

However, possibly due to lack of full understanding of your acticle, I find that the obtained angular distributions based on my code are not consistent with the other packages. The most confusing issue is that when I try to exchange the order of two daughter particles, the obtained amplitude of single event sometimes changes.

Thus, I hope to receive your kind help on them. Here is my questions:

Taking $\Lambda_c^+ \rightarrow pK^-\pi^+$ with isobar K^* as an example, in your acticle the indecies of $pK^-\pi^+$ are 1,2,3. When I re-order them as 2,1,3 by exchanging the index p, K^- , should I expect the same amplitude? Based on my code, I find most of part of Eq. 7 is exactly the same, but the $d_{\nu, \tau-\lambda'_k}^J(\hat{\theta}_{k(1)})$ in Eq 7 and $D_{\Lambda, \nu}^J(\alpha, \beta, \gamma)$ in Eq 3 changes. According to your explanation in

article, the $d_{\nu, \tau-\lambda'_k}^J(\hat{\theta}_{k(1)})$ is responsible for the different (α, β, γ) in different decay chains.

So I think them should be varied when I re-order the indecies. However, the numerical result of the obtained amplitude is not the same.

I fail to figure out the bug in my code. Could you please have a look on it? I attach an example in this email. Thanks in advance.